Note Title

Solution 1

1/25/2009

Prob. 5.1

a)
$$\left.\begin{aligned}\vec{r} &= \vec{r}_1 - \vec{r}_2 \\ \vec{R} &= \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1+m_2}\end{aligned}\right\} \Rightarrow \begin{cases}\vec{r}_1 = \vec{R} + \frac{m_2}{m_1+m_2}\vec{r} = \vec{R} + \frac{\mu}{m_1}\vec{r} \\ \vec{r}_2 = \vec{R} - \frac{\mu}{m_2}\vec{r}_2\end{cases}$$

where $\mu = \frac{m_1 m_2}{m_1+m_2}$

$$\nabla_1 = \frac{\partial}{\partial \vec{r}} \cdot \underbrace{\frac{\partial \vec{r}}{\partial \vec{r}_1}}_{=1} + \frac{\partial}{\partial \vec{R}} \underbrace{\frac{\partial \vec{R}}{\partial \vec{r}_1}}_{=\frac{m_1}{m_1+m_2}} = \nabla_{\vec{r}} + \frac{m_1}{m_1+m_2}\nabla_{\vec{R}} = \nabla_{\vec{r}} + \frac{\mu}{m_2}\nabla_{\vec{R}}$$

Similarly

$$\nabla_2 = \frac{\partial}{\partial \vec{r}} \frac{\partial \vec{r}}{\partial \vec{r}_2} + \frac{\partial}{\partial \vec{R}} \cdot \frac{\partial \vec{R}}{\partial \vec{r}_2} = -\nabla_{\vec{r}} + \frac{\mu}{m_1}\nabla_{\vec{R}}$$

b).
$$\left[-\frac{\hbar^2}{2m_1}\nabla_1^2 - \frac{\hbar^2}{2m_2}\nabla_2^2 + V(r_1 - r_2)\right]\psi = E\psi \qquad ①$$

$$\nabla_1^2 = \left(\nabla_{\vec{r}} + \frac{\mu}{m_2}\nabla_{\vec{R}}\right)^2 = \nabla_{\vec{r}}^2 + \frac{\mu^2}{m_2^2}\nabla_{\vec{R}}^2 + \frac{2\mu}{m_2}\nabla_{\vec{r}}\cdot\nabla_{\vec{R}}$$

$$\nabla_2^2 = \nabla_{\vec{r}}^2 + \frac{\mu^2}{m_1^2}\nabla_{\vec{R}}^2 - \frac{2\mu}{m_1}\nabla_{\vec{r}}\cdot\nabla_{\vec{R}}$$

The LHS of ① becomes

$$\begin{aligned}LHS &= \left(-\frac{\hbar^2}{2m_1}\nabla_{\vec{r}}^2 - \frac{\hbar^2}{2m_1}\frac{\mu^2}{m_2^2}\nabla_{\vec{R}}^2 - \cancel{\frac{\hbar^2}{2m_1}\cdot\frac{2\mu}{m_2}\nabla_{\vec{r}}\cdot\nabla_{\vec{R}}}\right. \\ &= \left.-\frac{\hbar^2}{2m_2}\nabla_{\vec{r}}^2 - \frac{\hbar^2}{2m_2}\frac{\mu^2}{m_1^2}\nabla_{\vec{R}}^2 - \cancel{\frac{\hbar^2}{2m_1}\frac{2\mu}{m_1}\nabla_{\vec{r}}\cdot\nabla_{\vec{R}}} + V(\vec{r})\right)\psi \\ &= \left[-\frac{\hbar^2}{2}\left(\frac{1}{m_1}+\frac{1}{m_2}\right)\nabla_{\vec{r}}^2 - \frac{\hbar^2}{2m_1m_2}\left(\frac{1}{m_2}+\frac{1}{m_1}\right)\mu^2\nabla_{\vec{R}}^2 + V(\vec{r})\right]\psi \\ &= \left[-\frac{\hbar^2}{2\mu}\nabla_{\vec{r}}^2 - \frac{\hbar^2}{2(m_1+m_2)}\nabla_{\vec{R}}^2 + V(\vec{r})\right]\psi\end{aligned}$$

c). let $\psi(\vec{R},\vec{r}) = \psi_R(\vec{R})\psi_r(\vec{r})$

$$\Rightarrow \left[-\frac{\hbar^2}{2\mu}\nabla_{\vec{r}}^2 - \frac{\hbar^2}{2(m_1+m_2)}\nabla_{\vec{R}}^2 + V(\vec{r})\right]\psi_R(\vec{R})\psi_r(\vec{r}) = E\,\psi_R(\vec{R})\psi_r(\vec{r})$$

Both sides divided by $\psi_R(\vec{R})\psi_r(\vec{r})$:

$$\underbrace{-\frac{\hbar^2}{2\mu}\frac{\nabla_r^2\psi_r(\vec{r})}{\psi_r(\vec{r})}}_{①} \underbrace{-\frac{\hbar^2}{2(m_1+m_2)}\frac{\nabla_R^2\psi_R(\vec{R})}{\psi_R(\vec{R})}}_{②} + \underbrace{V(\vec{r})}_{③} = E$$

① + ③ is only a function of $\vec{r}$, the rest terms do not contain $\vec{r}$. So ① + ③ must be a constant.

$$\Rightarrow \begin{cases} -\frac{\hbar^2}{2\mu}\frac{\nabla_r^2\psi_r(\vec{r})}{\psi_r(\vec{r})} + V(\vec{r}) = E_r \\ -\frac{\hbar^2}{2(m_1+m_2)}\frac{\nabla_R^2\psi_R(\vec{R})}{\psi_R(\vec{R})} = E_R \end{cases} \qquad E_r + E_R = E.$$

$$\Rightarrow \begin{cases} -\frac{\hbar^2}{2\mu}\nabla_r^2\psi_r(\vec{r}) + V(\vec{r})\psi_r(\vec{r}) = E_r\psi_r(\vec{r}) \\ -\frac{\hbar^2}{2(m_1+m_2)}\nabla_R^2\psi_R(\vec{R}) = E_R\psi_R(\vec{R}) \end{cases} \quad \text{w/} \quad E_r + E_R = E. \qquad \text{Q.E.D.}$$

Prob. 5.4

a) $\psi_\pm(\vec{r}_1,\vec{r}_2) = A[\psi_a(\vec{r}_1)\psi_b(\vec{r}_2) \pm \psi_b(\vec{r}_1)\psi_a(\vec{r}_2)]$

$$1 = \int|\psi_\pm(\vec{r}_1,\vec{r}_2)|^2 d\vec{r}_1\, d\vec{r}_2 = |A|^2\int d\vec{r}_1 d\vec{r}_2 \Big\{|\psi_a(\vec{r}_1)|^2|\psi_b(\vec{r}_2)|^2 + |\psi_b(\vec{r}_1)|^2|\psi_a(\vec{r}_2)|^2$$
$$\pm \psi_a^*(\vec{r}_1)\psi_b^*(\vec{r}_2)\psi_b(\vec{r}_1)\psi_a(\vec{r}_2) \pm \psi_a(\vec{r}_1)\psi_b(\vec{r}_2)\psi_b^*(\vec{r}_1)\psi_a^*(\vec{r}_2)\Big\}$$

$$= |A|^2\int d\vec{r}_1 \underbrace{|\psi_a(\vec{r}_1)|^2}_{1}\int d\vec{r}_2 \underbrace{|\psi_b(\vec{r}_2)|^2}_{1} + |A|^2\int d\vec{r}_1|\psi_b(\vec{r}_1)|^2\int d\vec{r}_2|\psi_a(\vec{r}_2)|^2$$
$$\pm|A|^2\int d\vec{r}_1\, \underbrace{\psi_a^*(\vec{r}_1)\psi_b(\vec{r}_1)}_{0}\int d\vec{r}_2\, \psi_b^*(\vec{r}_2)\psi_a(\vec{r}_2) + \text{c.c.}$$

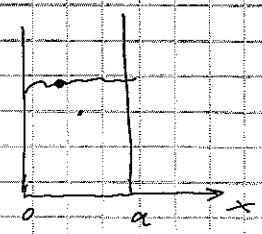
$$= 2|A|^2$$

$$\Rightarrow |A| = \frac{1}{\sqrt{2}}$$

b) $\psi_a = \psi_b$, $\psi_\pm(\vec{r}_1,\vec{r}_2) = 2A\,\psi_a(\vec{r}_1)\psi_a(\vec{r}_2)$

$$\Rightarrow 4A^2 = 1 \Rightarrow |A| = \frac{1}{2}$$

Prob 5.5



a) $H = -\frac{\hbar^2}{2m}\nabla_1^2 - \frac{\hbar^2}{2m}\nabla_2^2$

$$\begin{cases} H\psi = E\psi & (0 < x_1, x_2 < 0) \\ \psi = 0 & \text{else} \end{cases}$$

$\psi_n(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right)$, $\psi_{n_1n_2}(x_1,x_2) = \varphi_{n_1}(x_1)\varphi_{n_2}(x_2)$

$$H\psi_{n_1 n_2} \quad \left[+\frac{\hbar^2}{2m}\left(\frac{n_1\pi}{a}\right)^2 + \frac{\hbar^2}{2m}\left(\frac{n_2\pi}{a}\right)^2\right]\psi_{n_1 n_2}(x) = (n_1^2+n_2^2)\underbrace{\frac{\pi^2\hbar^2}{2ma^2}}_{\equiv K}\psi_{n_1 n_2}(x)$$

For the fermion ground state, obviously it is a eigen state w/ eigenvalue $5\frac{\pi^2\hbar^2}{2ma^2}$.

b). distinguishable: ψ_{22} & ψ_{13} or ψ_{31},

eigenvalue: $8K$, $10K$, $10K$.

identical bosons: $\psi_{(2,2)}(x_1,x_2) = \psi_{22}(x_1,x_2) \Rightarrow E = 8K$

$\psi_{(1,3)}(x_1,x_2) = \underbrace{\psi_{13}(x_1,x_2) + \psi_{13}(x_2,x_1)}_{\text{up to a normalization factor}} \Rightarrow E = 10K$

identical fermions: $\psi_{(1,3)}(x_1,x_2) = \psi_{13}(x_1,x_2) - \psi_{31}(x_1,x_2) \Rightarrow E = 10K$

$\psi_{(2,3)}(x_1,x_2) = \psi_{23}(x_1,x_2) - \psi_{32}(x_1,x_2) \Rightarrow E = 13K$